Shrimp Exports Forecast of Ecuador: A Comparative Analysis of Statistical Models for 2025

Delivering Reliable Monthly Production Insights for Strategic Planning

Author: Sebastián Arias F. January 2025

Executive Summary

Ecuador's shrimp industry plays a vital role in the global aquaculture market. Accurate forecasting of shrimp production is crucial for meeting international demand, optimizing logistics, and supporting economic planning. This white paper evaluates the performance of two statistical models—Exponential Smoothing (Holt-Winters), ARMA/ ARIMA—using monthly export data from January 2017 to December 2024. The study aims to provide reliable monthly shrimp export forecasts for 2025, offering actionable insights for stakeholders to optimize decision-making in production, processing factories and logistics.

Introduction

Ecuador's shrimp production and exports have established themselves as foundational to the nation's economy. The industry has significantly impacted job creation, export revenue generation, infrastructure investment, foreign exchange inflow, and economic growth. Sustainability initiatives and support for local communities further reinforce its strategic importance.

Despite its successes, the sector faces notable challenges, including global supply pressures, seasonal production variability, rising production costs, and industry consolidation. These factors underscore the importance of accurate data to inform export strategies, optimize production processes, and sustain competitive advantage.

This paper applies statistical models—Holt-Winters and ARMA/ARIMA—to analyze shrimp export data (2017–2024), generating reliable monthly forecasts for 2025. By capturing trends and seasonality, these forecasts aim to inform strategies for growth and stability in Ecuador's shrimp industry and its stakeholders.

Methodology

Data: Monthly shrimp export data (January 2017–December 2024), sourced from Statistic and Co.

Models:
Holt-Winters: Captures trends and seasonal patterns, making it suitable for time series with regular seasonality.
ARMA (Autoregressive Moving Average): Ideal for stationary data, combining autoregressive and moving average components.
ARIMA (Autoregressive Integrated Moving Average): Extends ARMA to handle non-stationary data through differencing, enabling it to model trends.

Evaluation Metrics:

- Mean Absolute Error (MAE).
- Root Mean Square Error (RMSE).
- Mean Absolute Percentage Error (MAPE).
- Others

Results and Discussion

Model 1: Exponential Smoothing (Holt-Winters)

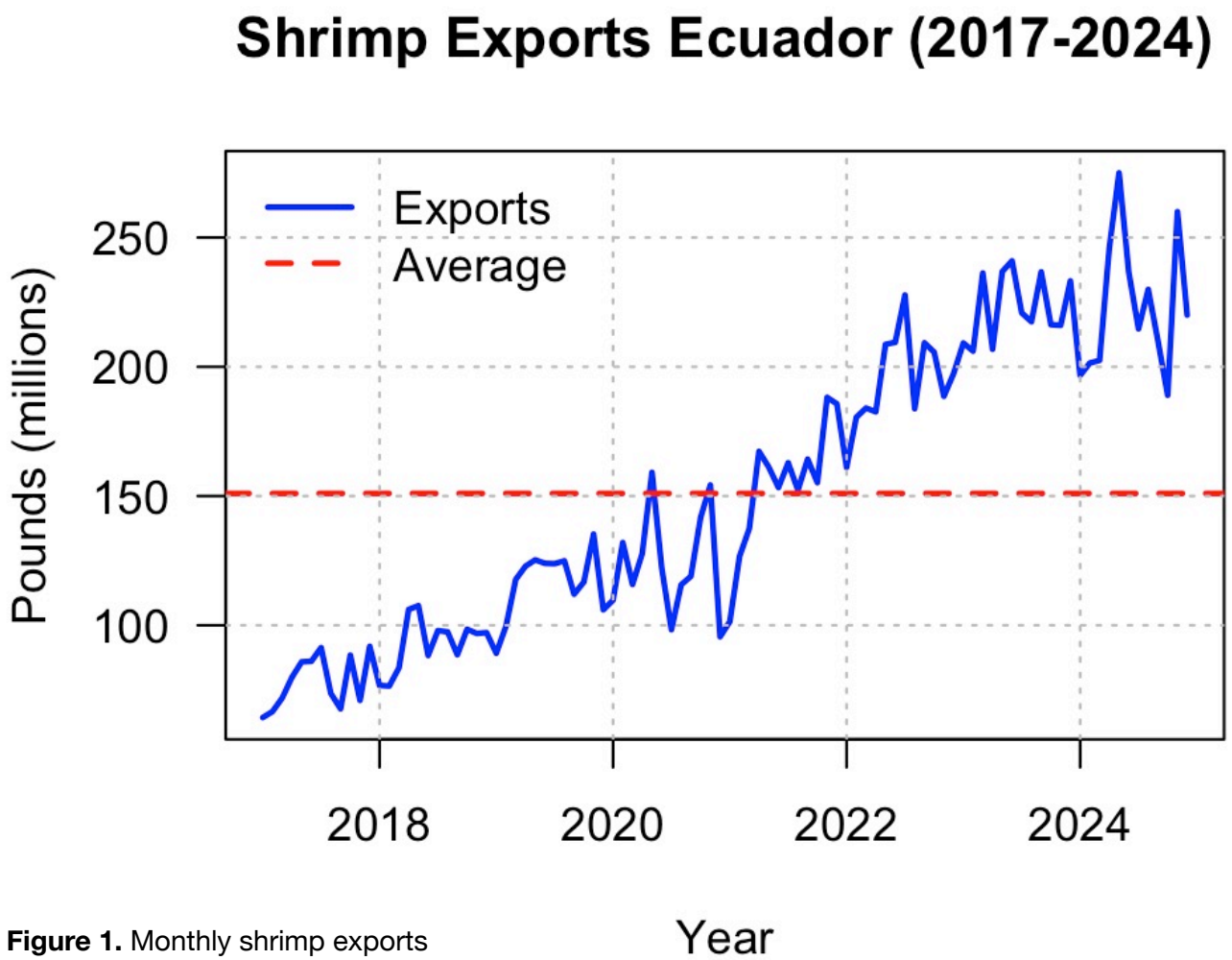

Figure 1. Monthly shrimp exports

The graph above illustrates Ecuador's shrimp exports from January 2017 to November 2024, measured in millions of pounds. The blue line represents the monthly export data, while the red dashed line indicates the average export volume over this period.

A clear upward trend in shrimp exports is observed over the analyzed period. Exports steadily increased from approximately 80 million pounds in early 2017 to peaks surpassing 250 million pounds in late 2024. This growth demonstrates the robust development of Ecuador's shrimp industry, driven by factors such as advancements in farming techniques, technology incorporation, export infrastructure, and increasing global demand.

Key observations include:

1. **Trend and Seasonality:** The data shows significant seasonality, with fluctuations around the mean (red dashed line). Seasonal peaks suggest that production aligns with specific periods, possibly driven by environmental conditions, stocking densities, harvested weights or market demand cycles.

2. **Rapid Growth in Recent Years:** Post-2020, the growth rate of exports accelerates, highlighting the effectiveness of recent industry improvements. This growth could also be linked to the recovery from global disruptions, such as the COVID-19 pandemic, increasing shrimp farm areas and more cycles per year.

3. **Volatility in Exports:** While the overall trend is positive, there are periodic sharp declines in exports. These drops may be attributed to challenges such as adverse weather, disease outbreaks, stocking densities or global market volatility.

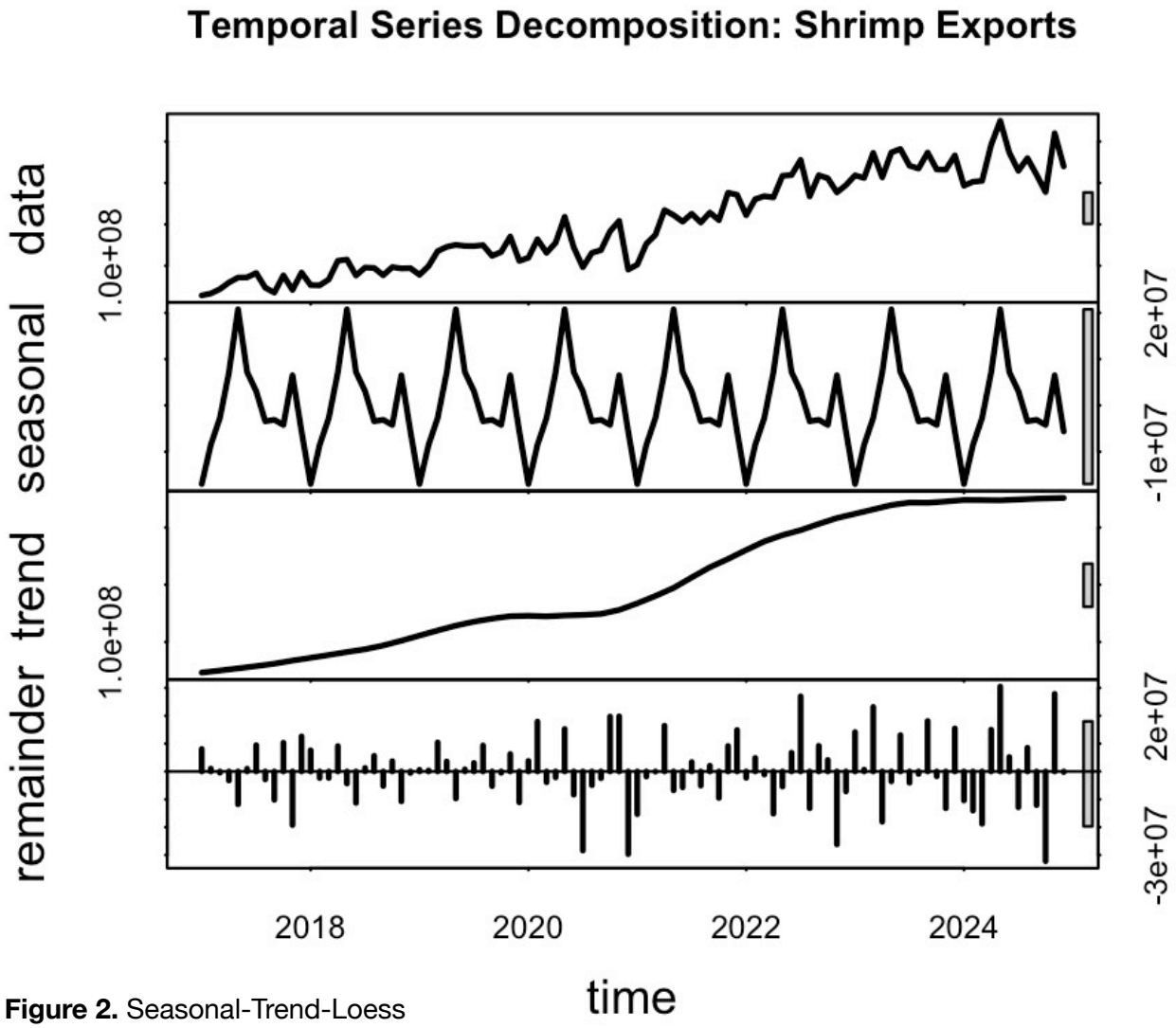

Figure 2. Seasonal-Trend-Loess

This graph provides a Seasonal-Trend-Loess (STL) decomposition of the shrimp export data, breaking the time series into its three primary components: original data, seasonal variation, trend, and residuals.

1. Data (Original Series): The top panel represents the raw shrimp export data from January 2017 to December 2024. A gradual increase in shrimp exports is observed, with noticeable seasonal fluctuations superimposed on the upward trend.

2. Seasonal (Seasonality): The second panel highlights the seasonal component, showing a recurring annual pattern in shrimp exports. Peaks consistently occur during specific months, indicating well-defined seasonality likely driven by favorable production cycles or market demand.

3. Trend (Tendency): The third panel displays the long-term trend component, illustrating sustained growth in shrimp exports throughout the analyzed period. This upward trajectory reflects increasing demand in international markets and improvements in Ecuador's shrimp production and export capacity.

4. Remainder (Residuals): The bottom panel captures the variability not explained by the trend or seasonality. The residuals are relatively small, suggesting that the majority of the variability in the time series is accounted for by the seasonal and trend components.

This decomposition provides critical insights into the dynamics of shrimp exports:

- The strong seasonal component underlines the need for export strategies that align with peak production periods to optimize market performance.
- The robust upward trend reflects the growing global importance of Ecuador's shrimp industry and indicates that the forecast for 2025 is likely to show continued growth.
- Minimal residual variability confirms that the models selected (Holt-Winters, ARMA) are well-suited for forecasting as they can effectively capture these patterns.

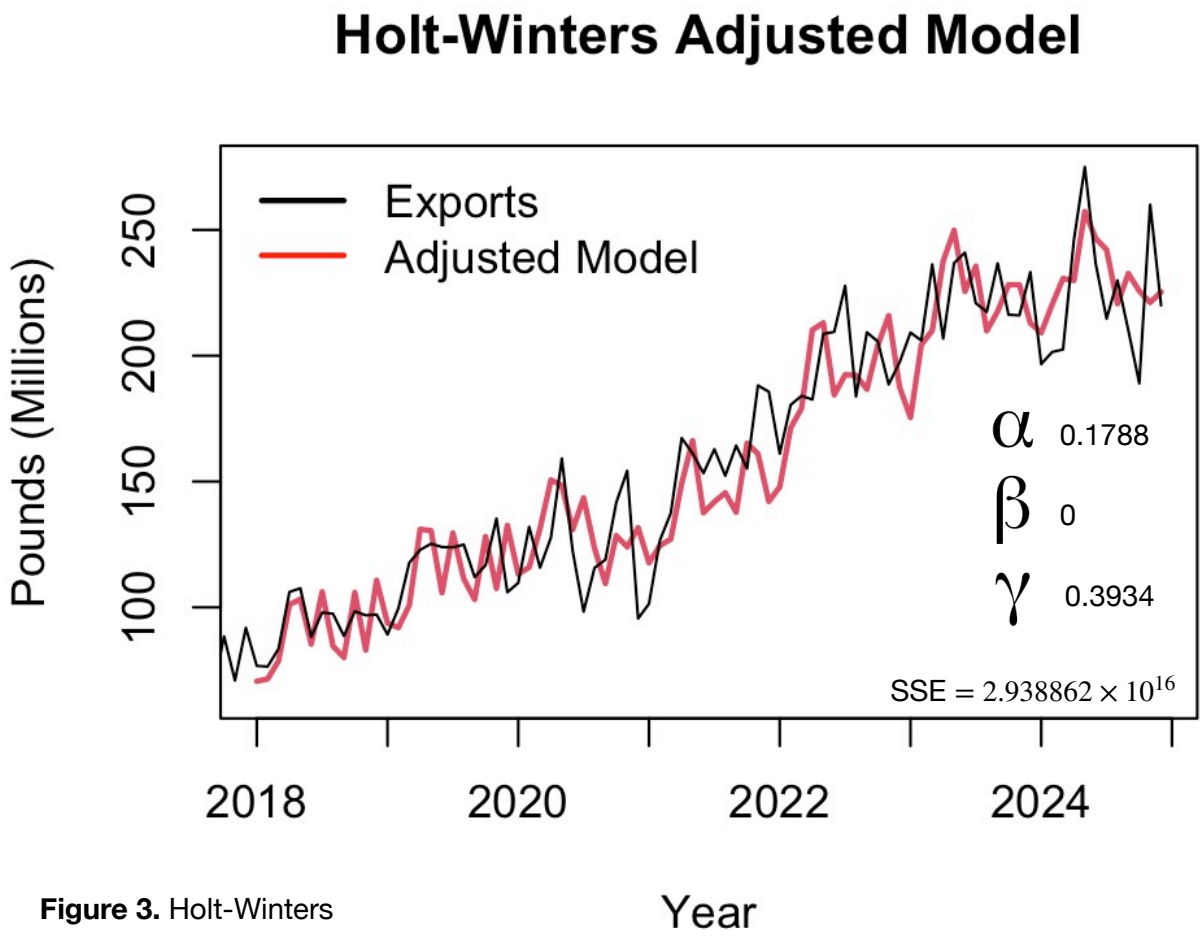

Figure 3. Holt-Winters

This graph compares the observed shrimp export values with the values adjusted by the Holt-Winters model:

Adjusted Model (Red Line):
- Captures seasonal peaks and the upward trend effectively.
- The model closely follows the observed values, indicating a good fit to the data.

Observed Values (Black Line):
- Represents the actual shrimp export data from 2017 to 2024.

- Differences between the observed and adjusted values are minimal, demonstrating a low error rate.

Model Components

- Alpha = 0.1788: Controls the smoothing of the level values. A low value indicates that the model places more weight on historical data rather than recent observations, which is suitable for datasets with stable long-term trends.

- Beta = 0: Controls the smoothing of the trend. A value of 0 suggests that the model did not detect a significant trend component, likely due to the dominance of the seasonal and level components in the data.

- Gamma = 0.3934: Controls the smoothing of the seasonality. An intermediate value indicates that seasonality is present and plays an important role in the model.

- Sum of Squared Errors (SSE): Represents the total accumulated error of the model. While the absolute value is large (due to the units of the data-millions), the focus is on minimizing this value when tuning the model. The relatively low SSE compared to the scale of the data suggests that the Holt-Winters model provides a good fit to the shrimp export dataset.

At this point Holt-Winters model is suitable for these data, as it effectively captures both the seasonality and the trend present in the time series. This makes it a reliable method for forecasting shrimp exports, ensuring accuracy in predicting seasonal fluctuations and long-term growth trends.

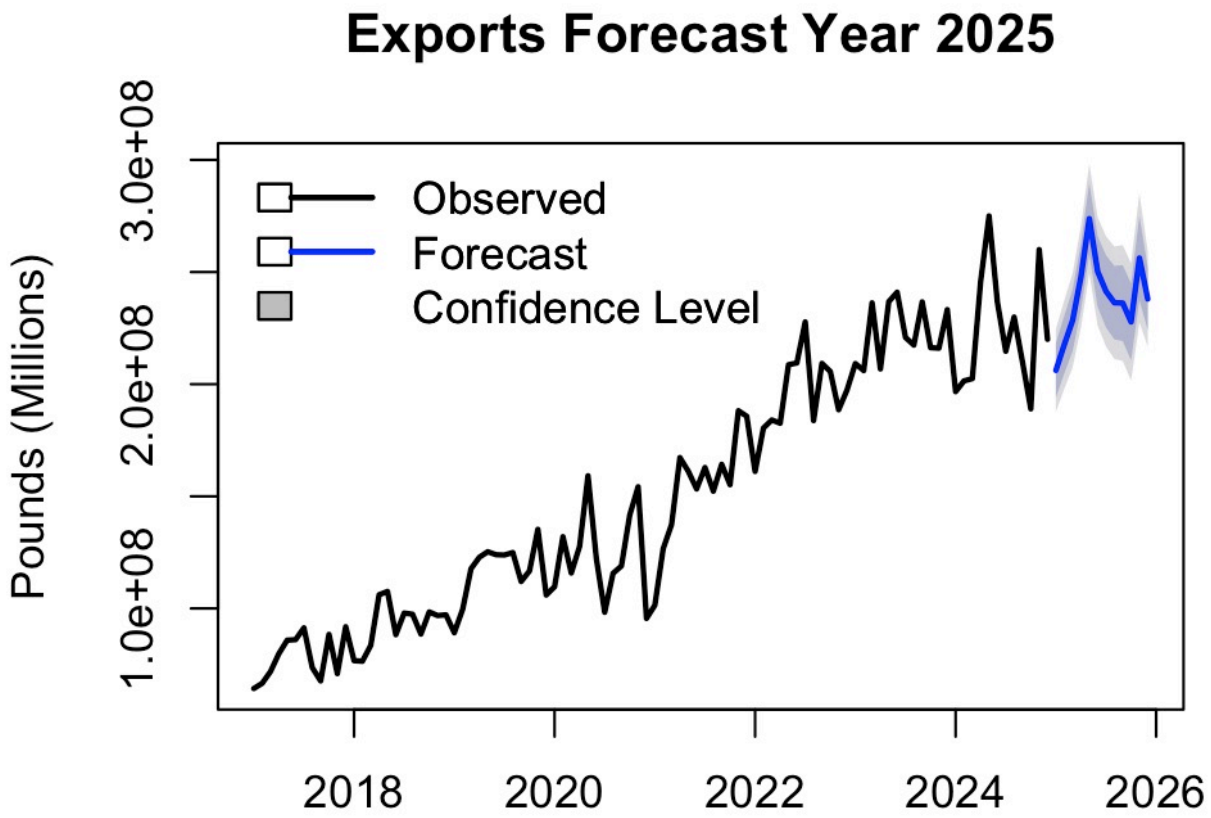

Figure 4. Holt-Winters Forecast Year 2025

The forecasted values represent the predicted shrimp exports for each month of 2025. These values indicate a continuation of the upward trend observed in previous years, reflecting increasing demand and export capacity.

These intervals highlight the model's reliability in capturing the possible variations in shrimp exports due to unforeseen market conditions or external factors.

Month / Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2025	206.166.613	194.126.848	218.206.377	187.753.382	224.579.843
Feb 2025	217.582.983	204.729.346	230.436.619	197.925.042	237.240.923
Mar 2025	228.405.236	214.727.676	242.082.797	207.487.212	249.323.260
Apr 2025	247.842.598	233.165.595	262.519.601	225.396.059	270.289.137
May 2025	273.774.787	257.894.845	289.654.729	249.488.511	298.061.063
Jun 2025	250.235.038	234.463.667	266.006.409	226.114.808	274.355.268
Jul 2025	241.392.929	225.349.224	257.436.633	216.856.200	265.929.658
Aug 2025	236.319.481	219.918.775	252.720.188	211.236.766	261.402.197
Sep 2025	236.171.966	219.270.657	253.073.275	210.323.644	262.020.288
Oct 2025	227.774.358	210.705.777	244.842.939	201.670.216	253.878.500
Nov 2025	256.227.683	237.486.624	274.968.741	227.565.707	284.889.658
Dec 2025	237.997.385	224.149.654	251.845.115	216.819.109	259.175.661

Table 1. Holt-Winters Forecast Year 2025 - Confidence Intervals

Conclusion of Holt-Winters Forecast Analysis

The 2025 forecast provides a robust projection of Ecuador's shrimp export trajectory, with confidence intervals offering insights into the expected range of monthly export volumes. The model's ability to capture both the seasonality and overall growth trends ensures that these forecasts are reliable for stakeholders in planning production, logistics, and policy measures. The inclusion of confidence intervals enhances the interpretability of the results, allowing stakeholders to prepare for potential variations in export volumes.

Model 2: ARMA (Autoregressive Moving Average)

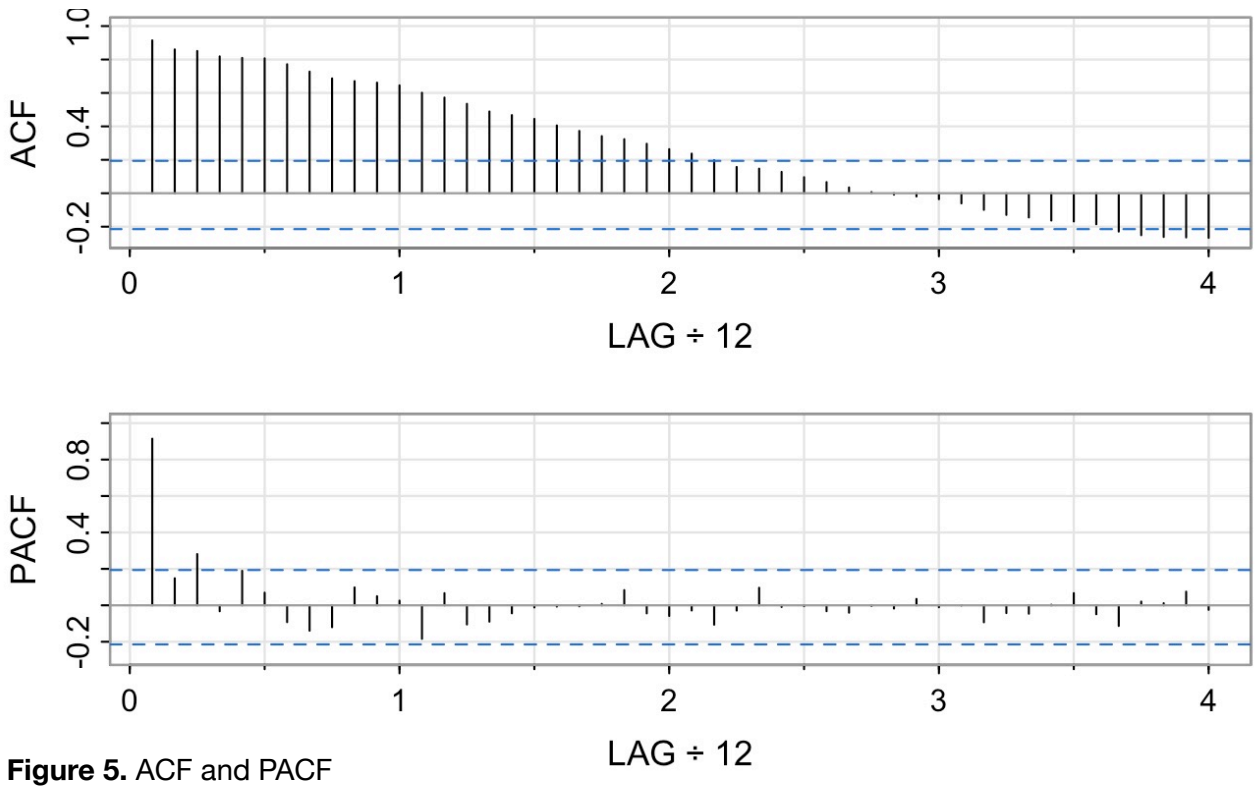

Figure 5. ACF and PACF

The Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots are crucial for determining the structure of the time series and selecting the appropriate ARMA model. The ACF plot (top panel) shows significant positive correlations over multiple lags, suggesting a strong persistence in shrimp export data, likely influenced by seasonality. The PACF plot (bottom panel) exhibits a gradual decline, indicating that a seasonal ARMA or ARIMA model might be suitable to capture the data's behavior effectively. However ARMA can be more suitable if data behave stationarity.

Stationarity Tests

To assess the stationarity of the time series, Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests were conducted:

1. Augmented Dickey-Fuller Test:
- Type 1 (No Drift, No Trend): The ADF statistic is not significant ($p > 0.05$), indicating non-stationarity.
- Type 2 (With Drift, No Trend): Similarly, the p -values are above the significance level, suggesting the presence of a unit root.
- Type 3 (With Drift and Trend): The ADF statistic becomes significant ($p < 0.01$) at lower lags, confirming stationarity when a trend component is included.

2. Phillips-Perron Test: Results are consistent with the ADF test. At higher lags, the p -values drop below 0.01 with drift and trend, indicating stationarity.

Conclusion on Stationarity

The tests suggest that the time series becomes stationary when adjusted for drift and trend, as seen in Type 3 results for both ADF and PP tests. This is essential for fitting ARMA models, as these methods require stationary data. The evidence from both ACF/PACF plots and stationarity tests supports the use of seasonal differencing and trend adjustment to enhance model accuracy.

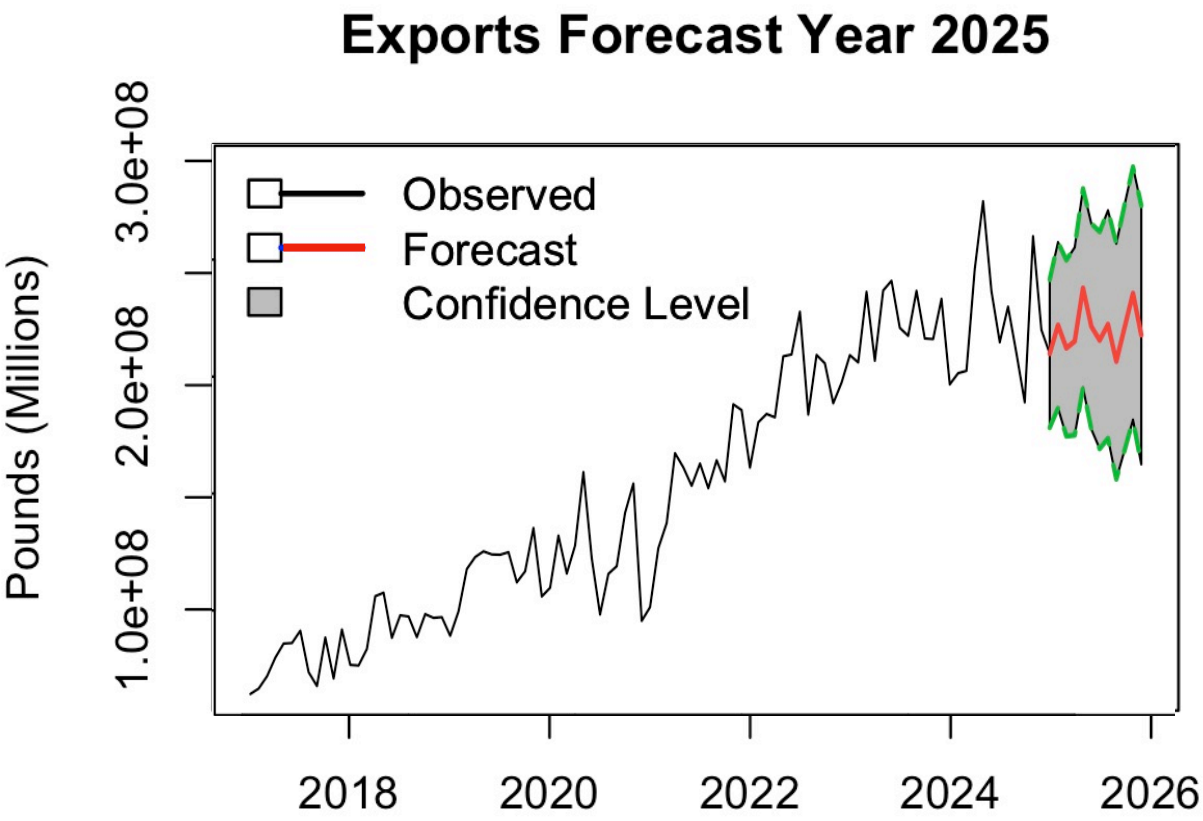

Figure 6. ARMA Model

The ARMA model (p=4, d=1, q=5) was fitted to the shrimp export data, providing the following insights:

1. Model Coefficients:

- AR (Auto-Regressive) Terms: Significant contributions from AR2 (-0.7874, SE = 0.0306) and AR4 (-0.9839, SE = 0.0176).
- MA (Moving Average) Terms: MA2 (0.6092, SE = 0.0526) and MA4 (0.8482, SE = 0.0804) showed strong influence.

These coefficients highlight complex autocorrelations captured by the model.

2. Model Performance:

- Akaike Information Criterion (AIC): 3449.06
- Bayesian Information Criterion (BIC): 3474.596
- Residual Variance: 2.381x10^14
- The relatively low AIC and BIC indicate the model's suitability for the data.
- Residual Mean: 3.65 x 10^6, close to zero, validating model accuracy.

3. Diagnostic Tests:

- Lilliefors Normality Test: p=0.5234, confirming the residuals normality.
- Box-Ljung Test: p=0.0097, indicating some autocorrelation in residuals.

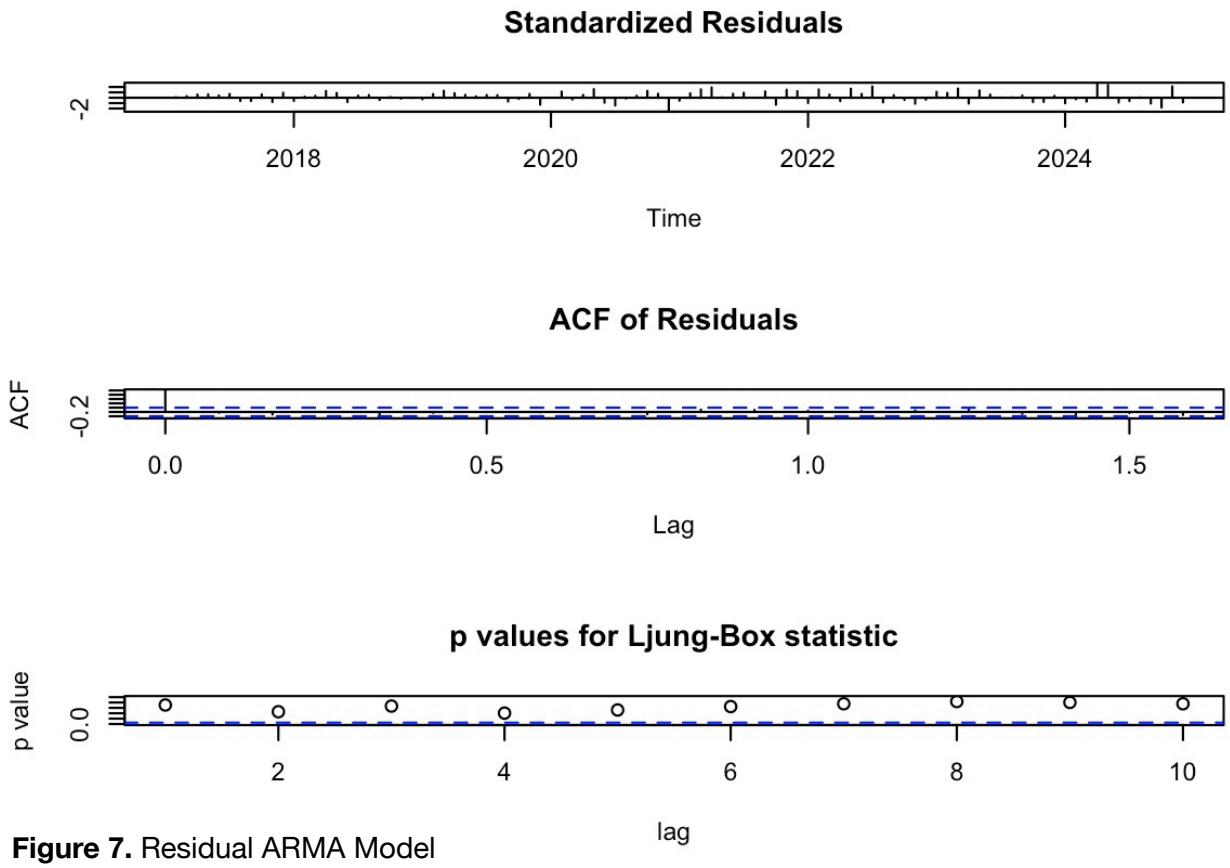

Figure 7. Residual ARMA Model

TEST	ARMA (0,1,3)	ARMA (4,1,5)	ARMA (5,1,5)	ARMA (6,1,5)
sigma^2	2,87E+17	2,52E+17	2,70E+17	2,67E+17
log likelihood	-1.714,49	-1.714,53	-1.715,74	-1.715,36
AIC	3.440,97	3.449,06	3.453,49	3.454,72
ME	4.293.501	3.657.890	3.646.955	3.760.126
RMSE	17.729.128	15.780.158	16.339.340	16.260.442
MAE	13.776.723	12.877.544	13.029.380	12.981.323
MPE	2,20	1,90	1,88	1,96
MAPE	9,38	8,92	8,89	8,87
MASE	0,89	0,83	0,84	0,84
ACF1	-0,08	-0,04	-0,05	-0,05
BIC	3.456,30	3.474,60	3.481,58	3.485,37
Normality	0,37	0,52	0,22	0,27
Box-Ljung	0,18	0,009	2,2E-16	0

Table 2. Comparison ARMA Models

The comparison between ARMA models with varying p, d, and q values reveals the trade-offs between model complexity and performance. The metrics provide insights into the behavior and suitability of each model for forecasting shrimp exports. ARMA (4,1,5) emerges as the most balanced model, demonstrating the best trade-off between accuracy, complexity, and residual behavior. While ARMA(0,1,3) is simpler and competitive in performance, ARMA(5,1,5) and ARMA(6,1,5) provide marginal improvements at the cost of complexity and poorer residual independence.

Month / Year	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2025	209.851.446	189.142.569	230.560.322	178.179.952	241.522.939
Feb 2025	222.019.310	198.840.003	245.198.617	186.569.619	257.469.001
Mar 2025	212.182.706	187.571.870	236.793.541	174.543.681	249.821.730
Apr 2025	215.103.692	188.854.595	241.352.790	174.959.162	255.248.222
May 2025	237.849.512	210.057.651	265.641.374	195.345.528	280.353.497
Jun 2025	221.450.883	192.697.951	250.203.814	177.477.069	265.424.697
Jul 2025	215.531.675	185.202.885	245.860.465	169.147.793	261.915.557
Aug 2025	222.348.574	190.483.963	254.213.186	173.615.857	271.081.292
Sep 2025	206.509.266	173.580.041	239.438.491	156.148.362	256.870.170
Oct 2025	221.144.876	186.805.616	255.484.136	168.627.510	273.662.242
Nov 2025	235.613.219	200.258.897	270.967.541	181.543.449	289.682.989
Dec 2025	217.821.124	181.649.255	253.992.993	162.501.024	273.141.224

Table 3. ARMA (4,1,5) Forecast Year 2025 - Confidence Intervals

Conclusion of ARMA Forecast Analysis

The ARMA(4,1,5) model effectively captures the dynamics of shrimp exports in Ecuador. While minor residual autocorrelations exist, the model's strong predictive capability is demonstrated by low residual errors and robust confidence intervals for forecasts. The predictions highlight a consistent upward trend with seasonal variations.

Comparative Analysis of Export Forecast Models:

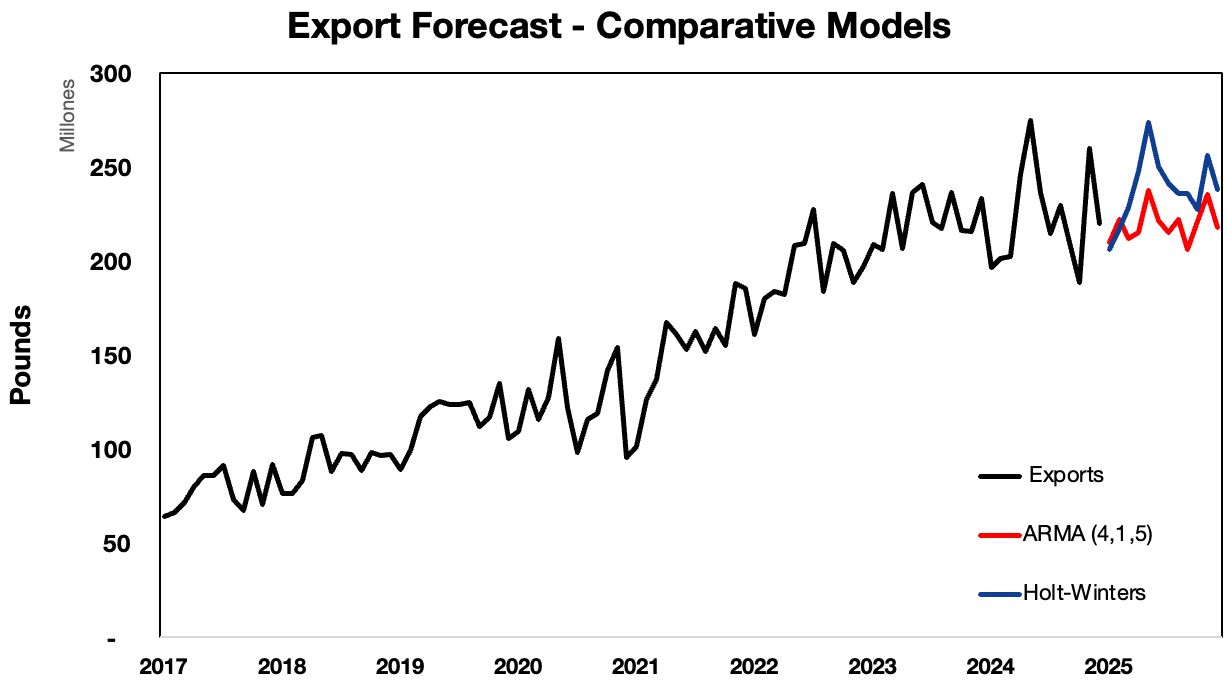

Figure 8. Comparative Models

The comparative graph demonstrates the performance of two forecasting models, Holt-Winters and ARMA(4,1,5), against the observed shrimp export data from 2017 to 2025. The analysis highlights the following key points:

1. Model Accuracy:

- Both models capture the upward trend in shrimp exports, reflecting the robust growth in production over the years.
- The Holt-Winters model aligns closely with the observed data, effectively capturing both the trend and seasonal variations.
- ARMA(4,1,5) also provides a strong forecast, but its predictions show slightly less alignment with the observed peaks and troughs compared to Holt-Winters.

2. Seasonality and Trend:

- The Holt-Winters model effectively incorporates seasonal patterns, as evidenced by its ability to replicate consistent peaks and dips across years.
- ARMA(4,1,5), while strong in overall trend prediction, appears to smooth some seasonal variations, leading to a less dynamic forecast.

3. Forecast for 2025:

The forecasted values for 2025 show a divergence in predictions between the two models

- Holt-Winters anticipates more pronounced fluctuations in exports, indicating a higher sensitivity to seasonal cycles.
- ARMA(4,1,5) predicts a steadier growth pattern with reduced variance.

Both models include confidence intervals (Fig. 4 and 6), providing a range of potential outcomes to account for forecast uncertainty.

Use Case Recommendation:

- For stakeholders prioritizing long-term planning and trend analysis, ARMA(4,1,5) offers a simpler and reliable forecast.
- For operational decisions requiring detailed understanding of seasonal dynamics, the Holt-Winters model is more suitable.

- For feed producers, monitoring stocking densities, FCR behavior, seasonal weather and harvested weight could be mandatory for better understanding its demand.
- This comparison underscores the importance of aligning model selection with specific forecasting objectives, balancing complexity, accuracy, and interpretability based on the shrimp export context.

Suggested Bibliography

- Hyndman, R. J., & Athanasopoulos, G. (2021). Forecasting: Principles and Practice (3rd ed.). OTexts. Retrieved from https://otexts.com/fpp3/
- FAO (2023). The State of World Fisheries and Aquaculture 2023. Food and Agriculture Organization of the United Nations. Retrieved from https://www.fao.org/
- Ecuadorian Ministry of Production, Foreign Trade, Investments and Fisheries (2023). Shrimp Export Statistics: 2017–2024.
- Box, G. E. P., Jenkins, G. M., & Reinsel, G. C. (2015). Time Series Analysis: Forecasting and Control (5th ed.). Wiley.